PRINCIPLES OF LOGIC

Logic

- study of reasoning
- specifically concerned with whether reasoning is correct
- focuses on the relationship among statements as opposed to the content of any particular statement.
- example:
  - All mathematicians wear sandals.
  - Anyone who wears sandals is an algebraist.
  - Therefore, all mathematicians are algebraists.

Technically, logic is of no help in determining whether any of these statements is true; however, if the first two statements are true, logic assures us that the last statement is also true.

Logical methods

- used in mathematics to prove theorems
- used in computer science to prove that programs do what they are alleged to do.

PROPOSITIONS

Truth value of a statement

- The only positive integers that divide 7 are 1 and 7 itself. True
- Gloria Macapagal Arroyo is the 14th president of the Philippines as she won in the 1998 presidential elections. False
- For every positive integer *n*, there is a prime number larger than *n*. True
- Earth is the only planet in the universe that has life.

  Truth value unknown at this time
- Buy two tickets to the rock concert for Friday. No truth value
- There is nothing on the television. ambiguous

Proposition

- a sentence that is either true or false, but not both
- typically expressed as a declarative sentence

Atomic Proposition

- propositions in which each one is an atomic entity
- truth or falsity is independent of any other propositions

Representation of proposition

we will use lowercase letters, such as p, q, and r, so that the notation

$$p: 1 + 1 = 3$$

defines p to be the proposition $1 + 1 = 3$.

Propositional Logic

- there are two possible truth values that can be associated with atomic propositions
- every proposition may be associated with only one of the truth values of **true** and **false**, at any given time
- sometimes determining the truth or falsity of a given proposition may not be possible but we know it must be true or false
- a proposition can never be both true and false at the same time although the truth value of a given proposition may change according to when it is stated.

Examples of Propositions

p: 1 + 1 = 2

q: Today is Tuesday.

r: Jack likes Jill.

s: Cedric is happy.

Three-valued Logic

- a proposition may have a third truth value of **undefined**
- it is quite difficult to attach a truth value to some propositions, thus, the undefined truth value for propositions in this field is accepted

However, in our study of discrete mathematics, we will tackle only two-valued propositions, and so we may conclude that a proposition is either true or false, we just have no means of determining precisely which.

NEGATION, CONJUNCTION AND DISJUNCTION

Propositions

- basic building blocks of any theory of logic
- just as natural numbers can be combined via addition or multiplication to form mathematical terms, so do atomic propositions via a variety of propositional logic operators to form compound propositions.

Compound Proposition

- a single proposition that results from the combination of atomic propositions

- example of atomic propositions:
  - It is raining.
  - I will take my umbrella.

- example of compound propositions:
  - It is raining and I will take my umbrella.
  - It is not raining and I will not take my umbrella.

Negation

Definition**.** The negation **of *p*,** denoted ~*p* or $\bar{p}$, is the proposition

not *p*.

Examples**:**

p: Eric is a vegetarian.

q: Today is Wednesday.

~*p*: Eric is not a vegetarian.

~*q*: Today is not Wednesday.

Law 2.1 (Properties of Negation)

1. (~ true) $\Leftrightarrow$ false

   (~ false) $\Leftrightarrow$ true

2. For any proposition *p*, the negation of the negation of *p* is *p* itself. That is,

   $(\sim \sim p) \Leftrightarrow p$

Negation is a unary logic operator because it takes in only one argument, that is, it operates only on one proposition – the proposition to be negated

Truth Table

- provides a means of representing the truth or falsity of logical statements
- the possible values of all atomic propositions contained in the proposition are listed, T for true and F for false, and the truth value of the overall proposition can thus be calculated for each possible combination

Truth Table : Negation

p	~p
T	F
F	T

Conjunction

Definition**:** Let p and q be propositions. The conjunction of p and q, denoted $p \wedge q$, is the proposition

p and q.

Remark: The proposition $p \wedge q$ can also be read as "p conjoined with q".

Example**:**

Given: p: $0 < 1$

q: $1 < 2$

$p \wedge q$: $(0 < 1) \wedge (1 < 2)$.

Truth Table : Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Notice that the conjuction $p \wedge q$ is true provided that p and q are both true; $p \wedge q$ is false otherwise.

Example of Conjunction

p: Eric is a vegetarian.

q: Eric eats hamburger.

- if p is false, and q is true, then the conjunction

  $p \wedge q$: Eric is a vegetarian and Eric eats hamburger is false.

- on the other hand, if Eric is indeed a vegetarian, so that p is true and q is false, then we obtain the same truth value of false for the conjunction $p \wedge q$.

Properties of Conjunction

For any propositions *p, q and r,*

1. Conjunction is idempotent, that is,
   $(p \wedge p) \Leftrightarrow p$

2. $(p \wedge \text{true}) \Leftrightarrow p$

3. $(p \wedge \text{false}) \Leftrightarrow \text{false}$

4. $(p \wedge \sim p) \Leftrightarrow \text{false}$

5. Conjunction is commutative, that is,
   $(p \wedge q) \Leftrightarrow (q \wedge p)$

6. Conjunction is associative, that is,
   $p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$

Truth Table : $p \wedge (\sim q)$

p	q	~q	$p \wedge \sim q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

Disjunction

- Definition**.** The disjunction of p and q, denoted $p \vee q$, is the proposition

  p or q

- Remark*: The proposition $p \vee q$ can also be read as "p disjoined with q".*

- Note: *Here, the or in the disjunction $p \vee q$ is used in the* ***inclusive*** *sense; that is, $p \vee q$ is considered to be true if either p or q or both are true, and $p \vee q$ is false only if both p and q are false. There is also an* ***exclusive-or*** *in which p exor q is true if either p or q, but not both, is true.*

Truth Table : *Disjunction*

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example of Disjunction

p: A decade is 10 years

q: A year is always 365 days

$p \vee q$: A decade is 10 years or a year is always 365 days

since p is true and q is false, then the disjunction **$p \vee q$ is true**

Example of Disjunction

p: $2^3 = 6$

q: 4 x 8 = 48

p $\vee$ q: 23 = 6 or 4 x 8 = 48

since *p* is false and *q* is false, then the disjunction **p $\vee$ q is false**

Properties of Disjunction

For any propositions p, q and r,

1. Disjunction is idempotent, that is,
   $(p \vee p) \Leftrightarrow p$
2. $(p \vee \text{false}) \Leftrightarrow p$
3. $(p \vee \text{true}) \Leftrightarrow \text{true}$
4. Disjunction is commutative, that is,
   $(p \vee q) \Leftrightarrow (q \vee p)$
5. Disjunction is associative, that is,

   $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$

Properties of Disjunction

For any propositions p, q and r,

6. For any proposition p

$$(\sim p \vee p) \Leftrightarrow \text{true}$$

7. Disjunction distributes through conjunction, that is, for any propositions p, q and r,

$$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$$

$$(p \wedge q) \vee r \Leftrightarrow (p \vee r) \wedge (q \vee r)$$

8. Conjunction distributes through disjunction, that is, for any propositions p, q and r,

$$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$$

$$(p \vee q) \wedge r \Leftrightarrow (p \wedge r) \vee (q \wedge r)$$

Exercise

Show that indeed disjunction distributes through conjunction, that is, for any propositions *p, q* and *r*,

$$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$$
$$(p \wedge q) \vee r \Leftrightarrow (p \vee r) \wedge (q \vee r)$$

Truth table: $p \vee (q \wedge r)$

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

CONDITIONAL PROPOSITIONS
AND
LOGICAL EQUIVALENCE

Conditional Propositions

Definition:

If *p* and *q* are propositions, the compound proposition

if *p* then *q*

is called a **conditional proposition** and is denoted

$$p \rightarrow q$$

Remark: The $\rightarrow$ operator is called the implication operator and the proposition *p* is called the hypothesis (or antecedent) and the proposition *q* is called the conclusion (or consequent).

Example: Conditional Propositions

If the Mathematics Department gets an additional budget, then it will hire one new faculty member.

p: If the Mathematics Department gets an additional budget

q: It will hire one new faculty member

If the Mathematics Department gets an additional budget, then it will hire one new faculty member.

Other Forms of $p \rightarrow q$

- Mary will be a good student <u>if</u> she studies hard.
- John may take calculus <u>only if</u> he has sophomore, junior, senior standing.
- <u>When</u> you sing, my ears hurt.
- A <u>necessary condition</u> for the Cubs to win the World Series is that they sign a right-handed relief pitcher.
- A <u>sufficient condition</u> for Ralph to visit California is that he goes to Disneyland.

Other Forms of $p \rightarrow q$

- The hypothesis is the clause following *if*
- The *only if* clause is the conclusion; that is, if *p* then *q* is considered logically the same as *p* only if *q*
- The "if *p* then *q*" formulation emphasizes the hypothesis, whereas the "*p* only if *q*" formulation emphasizes the conclusion; the difference is only the style.
- *When* means the same as *if*
- The conclusion expresses a necessary condition
- The hypothesis expresses a sufficient condition

Other Forms of $p \rightarrow q$

- Mary will be a good student <u>if</u> she studies hard.

  If Mary studies hard, then she will be a good student.

- John may take calculus <u>only if</u> he has sophomore, junior, senior standing.

  If John takes calculus then he has sophomore, junior, or senior standing.

- <u>When</u> you sing, my ears hurt.

  If you sing, then my ears hurt.

Other Forms of $p \rightarrow q$

- A necessary condition for the Cubs to win the World Series is that they sign a right-handed relief pitcher.

  If the Cubs win the World Series, then they sign a right-handed relief pitcher.

- A sufficient condition for Ralph to visit California is that he goes to Disneyland.

  If Ralph goes to Disneyland, then he visits California.

Truth Table: $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Note that the truth table for a conditional position shows a truth value of false only when the hypothesis is true and the conclusion is false. At all other cases, the statement is true.

Example: *conditional proposition*

p: $1 > 2$
q: $4 < 8$

$p \rightarrow q$ is true

$q \rightarrow p$ is false

Example: *conditional proposition*

Given: *p – true,* *q – false,* *r – true*

Find the truth values of the following propositions:

$$(p \wedge q) \rightarrow r$$

Relationship of propositions

- In ordinary language, the hypothesis and conclusion in a conditional proposition are normally related, but in logic, the hypothesis and conclusion in a conditional proposition are not required to refer to the same subject matter.

- For example, in logic, we permit propositions such as:

  If $5 < 3$, then Nelson Rockefeller was the president of the United States.

Relationship of propositions

- Logic is concerned with the form of propositions and the relation of propositions to each other and not with the subject matter itself.

- In fact, the given statement in the previous slide is true since the hypothesis is false.

- Notice that a true conditional proposition is different from a conditional proposition with a true conclusion.

Converse of a proposition

Recall:

***p*: 1 > 2** $\quad$ p $\rightarrow$ q is true $\quad$ q $\rightarrow$ p is false

***q*: 4 < 8**

- The above example shows that the proposition $p \rightarrow q$ can be true while the proposition $q \rightarrow p$ is false.
- We call the proposition $q \rightarrow p$ the converse of the proposition $p \rightarrow q$.
- Hence, a conditional proposition can be true while its converse is false.
- This shows that implication, unlike other propositional logical operators, is not commutative. Furthermore, it not idempotent and associative.

Example: *conditional proposition*

Given the conditional proposition

If $1 < 2$, then $3 < 6$.

Write the proposition symbolically. Write the converse of this statement symbolically and in words. Also, find the truth value of the conditional proposition and its converse.

Example: *conditional proposition*

Solution:

Let p: $1 < 2$, q: $3 < 6$.

- The given statement may be written symbolically as $p \rightarrow q$
- Since p and q are both true, this statement is true.
- The converse may be written symbolically as $q \rightarrow p$ and in words as

  If $3 < 6$, then $1 < 2$.
- Since p and q are both true, the converse is true.

LOGICAL EQUIVALENCE

Logical Equivalence

Definition: If p and q are propositions, the compound proposition

p if and only if q

is called a **biconditional proposition** and is denoted

$$p \leftrightarrow q$$

Remarks:

- The $\leftrightarrow$ operator is called the equivalence operator.
- An alternative way to state "p if and only if q" is "p is a necessary and sufficient condition for q."
- "p if and only if q" can also be written as "p iff q."

Truth Table: Logical Equivalence

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example: Logical Equivalence

p: $1 < 5$
q: $2 < 6$

$p \leftrightarrow q$ is true since both p and q are true

An alternative way to state this statement is:

A necessary and sufficient condition for $1 < 5$ is that $2 < 6$.

Property of Logical Equivalence

(Law 2.4) For any propositions p and q,

$$(p \to q) \leftrightarrow ((\sim p) \vee q).$$

Example:

p: It will rain this afternoon.
q: The Garcias will stay in.

$p \to q$: If it rains this afternoon, then the Garcias will stay in

$(\sim p) \vee q$: Either it will not rain this afternoon or the Garcias will stay in.

$$(p \to q) \leftrightarrow ((\sim p) \vee q)$$

Logically Equivalent Compound Propositions

Definition. Suppose that the compound propositions P and Q are made up of the propositions $p_1, \ldots, p_n$. We say that P and Q are **logically equivalent** and write

$$P \equiv Q$$

provided that given any truth values of $p_1, \ldots, p_n$, either P and Q are both true or P and Q are both false.

De Morgan's Laws (Law 2.5)

- For any propositions *p* and *q*,

$$\sim (p \vee q) \leftrightarrow ((\sim p) \wedge (\sim q))$$

$$\sim (p \wedge q) \leftrightarrow ((\sim p) \vee (\sim q)).$$

- These laws can also written as follows:

$$\overline{p \vee q} \equiv \overline{p} \wedge \overline{q}$$

$$\overline{p \wedge q} \equiv \overline{p} \vee \overline{q}$$

Proof of De Morgan's Law (i)

p	q	$\sim (p \vee q)$	$\sim p \wedge \sim q$
T	T	F	F
T	F	F	F
F	T	F	F
F	F	T	T

$$\overline{p \to q} \Leftrightarrow p \wedge \overline{q}$$

p	q	$\overline{p \to q}$	$p \wedge \overline{q}$
T	T	F	F
T	T	F	F
F	T	F	F
F	F	F	F

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

Properties of Equivalence (Law 2.6)

1. Equivalence is associative, that is, for any propositions *p, q* and *r*,

$$p \leftrightarrow (q \leftrightarrow r) \equiv (p \leftrightarrow q) \leftrightarrow r$$

2. Equivalence is commutative, that is, for any propositions *p* and *q*,

$$(p \leftrightarrow q) \equiv (q \leftrightarrow p)$$

3. Any proposition is equivalent to itself, that is, for any proposition *p*

$$(p \leftrightarrow p) \leftrightarrow true$$

Properties of Equivalence (Law 2.6)

4. No proposition is equivalent to its negation, that is, for any proposition *p*

$$(p \leftrightarrow (\sim p)) \leftrightarrow false$$

5. To say that one proposition *p* is equivalent to another proposition *q*, is the same as saying *p* implies *q*, and *q* implies *p*. That is, for any propositions *p* and *q,*

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p).$$

Contrapositive

Definition: The **contrapositive** (or **transposition**) of the conditional proposition $p \rightarrow q$ is the proposition $\sim q \rightarrow \sim p$

Remarks:

1. Notice the difference between the contrapositive and the converse. The converse of a conditional proposition merely reverses the roles of *p* and *q*, whereas the contrapositive reverses the roles of *p* and *q* and negates each of them.
2. The contrapositive is an alternative, logically equivalent form of the conditional proposition.

Example Contrapositive

Write the proposition

If $1 < 4$, then $5 > 8$

symbolically and write the converse and the contrapositive both symbolically and in words. Find the truth value of each proposition.

Answer:

Let p: $1 < 4$ q: $5 > 8$,

then the given proposition may be written symbolically as

$p \rightarrow q.$

The converse is $q \rightarrow p$ or in words If $5 > 8$, then $1 < 4$.

The contrapositive is $\sim q \rightarrow \sim p$ or in words,

If 5 is not greater than 8, then 1 is not less than 4.

We see that $p \rightarrow q$ is false, $q \rightarrow p$ is true, and $\sim q \rightarrow \sim p$ is false.

A theorem on contrapositive

The conditional proposition $p \rightarrow q$ and its contrapositive $\sim p \rightarrow \sim q$ are logically equivalent.

Proof:

p	q	$p \rightarrow q$	$\sim q \rightarrow \sim p$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Hierarchy of Logical Operators

1. negation (~) has the highest priority
2. conjunction ($\wedge$)
3. disjunction ($\vee$)
4. implication ($\rightarrow$)
5. logical equivalence ($\leftrightarrow$)

Example: Hierarchy of Logical Operators

- ✓ The proposition $\sim p \wedge q$ means $(\sim p) \wedge q$ rather than $\sim (p \wedge q)$.
- ✓ The proposition $p \vee q \rightarrow r$ means $(p \vee q) \rightarrow r$ rather than $p \vee (q \rightarrow r)$.

TAUTOLOGIES, CONTRADICTIONS AND CONTINGENCIES

Tautology

Definition**:**

A proposition which is always logically equivalent to *true,* no matter what the values of its constituent atomic propositions are, is called a tautology.

Tautology

Remark:

An easy method of determining whether or not a proposition is a tautology is to substitute truth values for the atomic propositions. If the truth value of the proposition is *true* for all combinations of values attributed to the atomic propositions, then, we may conclude that it is a tautology.

Example of Tautology

- $p \vee \sim p$
- $p \rightarrow p$
- $(p \wedge q) \vee (p \wedge \sim q) \vee (\sim p \wedge q) \vee (\sim p \wedge \sim q)$

Is $(p \vee \sim p) \rightarrow q$ a tautology?

Answer:

From the previous slide, we have seen that $p \vee \sim p$ is a tautology. If q is *false*, then $(p \vee \sim p) \rightarrow q$ is false. Hence $(p \vee \sim p) \rightarrow q$ is not a tautology.

Contradiction

Definition**:**

A proposition which is always *false* is called contradiction.

Example**:**

The proposition $\sim (p \rightarrow p)$ is a contradiction. If p is *true*, then the overall proposition is *false*, whereas if p is *false* the overall proposition is also *false*.

Contingency

Definition:

If a proposition is neither a tautology nor a contradiction, then it is called a contingency.

Contingency

Example:

Consider the proposition $p \rightarrow q$. If p is *true* and q is *false*, then the implication is *false*, whereas if p and q are both *true* the implication is *true*. Hence, the implication $p \rightarrow q$ is a contingency.

EQUATIONAL REASONING

Equational Reasoning

We can determine the truth or falsity of propositions by substituting one logically equivalent proposition for another.

Equational Reasoning

When conducting this process, we have to strictly apply the laws of propositional logic at each step.

This method of analysis is called Equational Reasoning.

Equational Reasoning

Problem: Prove that $p \rightarrow p$ is equivalent to *true* using the laws of propositional logic.

Proof:

$$
\begin{aligned}
& p \rightarrow p \\
\leftrightarrow \quad & \sim p \vee p && [\text{Law 2.4}] \\
\leftrightarrow \quad & \mathit{true} && [\text{Law 2.3.6}]
\end{aligned}
$$

Equational Reasoning

Problem: Prove that $p \to (p \to q)$ is logically equivalent to $(p \to q)$.

Proof:

$p \to (p \to q)$

$\leftrightarrow \quad \sim p \vee (p \to q)$ [Law 2.4]

$\leftrightarrow \quad \sim p \vee (\sim p \vee q)$ [Law 2.4]

$\leftrightarrow \quad (\sim p \vee \sim p) \vee q$ [Law 2.3.5)]

$\leftrightarrow \quad \sim p \vee q$ [Law 2.3.1 (Idemp of V]

$\leftrightarrow \quad (p \to q)$ [Law 2.4]

Equational Reasoning

Problem: Using equational reasoning, find the truth value of the proposition

$$(p \wedge q) \leftrightarrow (\sim p \vee \sim q).$$

Solution:

$(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$

$\leftrightarrow (p \wedge q) \leftrightarrow \sim(p \wedge q)$ [De Morgan's Law]

$\leftrightarrow$ *false* [Law 2.6.4

(No prop. is equiv. to its negation)]